

5. (a) Let  $H = (V, E)$  be a  $k$ -uniform hypergraph with  $|E| = m \leq \frac{r^{k-1}}{(r-1)^k}$ . Let the set of  $r$  color be  $[r]$ ,  $r \geq 2$ . Since we color the vertices of  $V$  uniformly at random, for any vertex  $v \in V$  and color  $c \in [r]$ ,  $P(v \text{ is colored } c) = \frac{1}{r}$ .  
 $\Rightarrow P(v \text{ is not colored } c) = \frac{r-1}{r}$ .

Also, since we color the vertices of  $V$  independently, for any edges  $e \in E$ ,  $P(e \text{ contains no vertex of color } c) = \left(\frac{r-1}{r}\right)^k$ . Let  $A_e$  be the event that  $e$  does not contain all the color, and  $A_{e,c}$  be the event that  $e$  is missing color  $c$ .  
 $\Rightarrow P(A_e) = P\left(\bigcup_{c=1}^r A_{e,c}\right) \leq \sum_{c=1}^r P(A_{e,c}) = r \cdot \left(\frac{r-1}{r}\right)^k = \frac{(r-1)^k}{r^{k-1}}$ .

(Note that  $P\left(\bigcup_{c=1}^r A_{e,c}\right) \neq \sum_{c=1}^r P(A_{e,c})$  since the events  $A_{e,c}$  and  $A_{e,c'}$  are not disjoint for any colors  $c, c'$ )

$\Rightarrow P(H \text{ is not rainbow-}r\text{-colorable})$   
 $= P\left(\bigcup_{e \in E} A_e\right) \leq \sum_{e \in E} P(A_e) = |E| \times P(A_e) < \frac{r^{k-1}}{(r-1)^k} \times \frac{(r-1)^k}{r^{k-1}} = 1$   
 $\Rightarrow P(H \text{ is rainbow-}r\text{-colorable}) > 0$

$\Rightarrow$  There exists some valid rainbow- $r$ -colorings for  $H$ .

$\Rightarrow H$  is rainbow- $r$ -colorable.

(b) Let  $V$  be a set of  $n$  vertices, where  $n$  will be chosen later.

We construct a random  $k$ -uniform hypergraph  $H = (V, E)$  by choosing  $m$  edges independently and uniformly at random from  $\binom{V}{k}$ . If we can show that  $P(H \text{ is rainbow } r\text{-colorable}) < 1$ , then there must exist a  $k$ -uniform hypergraph with at most  $m$  edges that is not rainbow  $r$ -colorable.

Fix an arbitrary  $r$ -coloring of the vertices  $\chi: V \rightarrow [r]$ .

Let  $n_1, n_2, \dots, n_r$  denote the sizes of the color classes under  $\chi$ , so  $\sum_{i=1}^r n_i = n$ .

A uniformly chosen edge  $e \in \binom{V}{k}$  fail to be rainbow under  $\chi$  if it misses at least one color. By the pigeonhole principle, there is some color class of size at least  $\lceil \frac{n}{r} \rceil$ .

Since  $\chi \mapsto \binom{\chi}{k}$  is convex and increasing, the probability that a random edge misses a color is minimized when the coloring is balanced. That is

$$p := \mathbb{P}(e \text{ is not rainbow under } \chi) \geq \frac{\binom{\lfloor \frac{r-1}{r}n \rfloor}{k}}{\binom{n}{k}} = \prod_{i=0}^{k-1} \frac{\lfloor \frac{r-1}{r}n \rfloor - i}{n - i}.$$

(Let  $n^*$  be the color whose color class of size at least  $\lceil \frac{n}{r} \rceil$ .  $n - \lceil \frac{n}{r} \rceil = \lfloor \frac{r-1}{r}n \rfloor$ .  $\binom{\lfloor \frac{r-1}{r}n \rfloor}{k}$  forms the set of edges missing color  $n^*$ , so for  $e \in \binom{\lfloor \frac{r-1}{r}n \rfloor}{k}$ ,  $e$  is not rainbow)

For  $n \gg k$ , using  $1 - x \geq e^{-x - O(x^2)}$ , we have

$$p = \left(\frac{r-1}{r}\right)^k \cdot \prod_{i=0}^{k-1} \left(1 - \frac{i}{(r-1)n} + O\left(\frac{i^2}{n^2}\right)\right) \\ \geq \left(\frac{r-1}{r}\right)^k \cdot \prod_{i=0}^{k-1} \exp\left(-\frac{i}{(r-1)n} + O\left(\frac{i^2}{n^2}\right)\right) = \left(\frac{r-1}{r}\right)^k \exp\left(-\frac{k^2}{2(r-1)n} + O\left(\frac{k^3}{n^2}\right)\right).$$

For this fixed coloring  $\chi$ , the probability that all  $m$  randomly chosen edges are rainbow is at most  $(1-p)^m \leq e^{-pm}$ .

Let  $\mathcal{A}_\chi$  be the event that  $H$  is rainbow colored by  $\chi$ .

$\Rightarrow \mathbb{P}(H \text{ is rainbow-}r\text{-colorable})$

$$= \mathbb{P}\left(\bigcup_{\chi \in [r]^V} \mathcal{A}_\chi\right) \leq \sum_{\chi \in [r]^V} \mathbb{P}(\mathcal{A}_\chi) = r^n \cdot (1-p)^m \leq \exp(n \ln r - pm).$$

This probability is strictly less than 1 whenever  $m > \frac{n \ln r}{p}$ .

Setting  $n = \frac{k^2}{2(r-1)}$ , we have

$$\frac{n \ln r}{p} \leq \frac{k^2}{2(r-1)} \cdot \ln r \cdot \left(\frac{r}{r-1}\right)^k \cdot \exp\left(\frac{k^2}{2(r-1)n} - O\left(\frac{k^3}{n^2}\right)\right) \\ = \frac{k^2}{2(r-1)} \cdot \ln r \cdot \left(\frac{r}{r-1}\right)^k \cdot \exp\left(1 - O\left(\frac{k^3}{n^2}\right)\right) \leq \frac{e \ln r}{2(r-1)} \cdot k^2 \cdot \left(\frac{r}{r-1}\right)^k.$$

Let  $C(r) = \frac{e \ln r}{2(r-1)}$ , which only depends on  $r$ .

Taking  $m = C(r) k^2 \left(\frac{r}{r-1}\right)^k$  ensures that  $\mathbb{P}(H \text{ is rainbow-}r\text{-colorable}) < 1$ , which implies that there exists some  $H$  not rainbow- $r$ -colorable.